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Efficient connectivity and intrinsic noise separation in recurrent spiking neural networks trained with e-prop

Davide Noè^{1,2,*}, Hideaki Yamamoto^{1,2,*} , Yuichi Katori^{3,4} and Shigeo Sato^{1,2} ¹ Research Institute of Electrical Communication, Tohoku University, Sendai, Japan² Graduate School of Engineering, Tohoku University, Sendai, Japan³ School of Systems Information Science, Future University Hakodate, Hakodate, Japan⁴ International Research Center for Neurointelligence (WPI-IRCN), The University of Tokyo, Tokyo, Japan

* Authors to whom any correspondence should be addressed.

E-mail: noe.davide.p5@dc.tohoku.ac.jp and hideaki.yamamoto.e3@tohoku.ac.jp**Keywords:** recurrent neural networks, spiking neurons, eligibility propagation, biologically plausible learning, computational neuroscienceSupplementary material for this article is available [online](#)

Abstract

Biologically plausible learning rules for neural networks, such as e-prop (eligibility propagation), are essential both for advancing neuromorphic computing and for understanding fundamental mechanisms of learning in animal brains. However, their behavior under different network conditions remains unclear. Here, we investigate the performance of the e-prop learning algorithm in recurrent spiking neural networks (RSNNs) across different levels of recurrent connectivity and input noise using a complex temporal credit assignment task, a supervised classification problem known to be solvable by rodents. We show that increased sparsity in the recurrent layer significantly enhances learning performance by promoting the generation of more diverse activation patterns. Analysis of the network's evolution further reveals that the e-prop-trained input layer evolves to route distinct inputs to different regions of the recurrent layer while suppressing the contribution of noise. This partially resembles signal routing functions attributed to the thalamus in mammalian sensory systems, providing additional support for the biological plausibility of e-prop. These findings offer promising insights for efficiency and advantages of biologically inspired training in RSNNs.

1. Introduction

With the burgeoning diffusion of machine learning [1, 2] in diverse fields such as healthcare [3, 4], software engineering [5], and renewable energy research [6], the amount of data and power required to train and operate artificial neural network models is increasing rapidly [7]. This trend has renewed our attention to the initial inspiration for artificial intelligence: the brain [8, 9]. Indeed, biological brains can learn with limited data [10] and exhibit remarkable power efficiency. For example, current large language models require several thousands of GPUs for training and inference, with individual units reaching up to 400 W as of this writing [11], causing a steep growth in energy demand [12, 13]. Meanwhile, the human brain continually operates on around 20 W [14]. These advantages sparked great interest in neuromorphic engineering, leading to the development of brain-inspired hardware such as Intel's Loihi [15] or our analog VLSI chip [16], which implement models that resemble biological brains: recurrent spiking neural networks (RSNNs) [17, 18].

RSNNs have been the subject of studies throughout the years, but researchers have struggled to overcome the underlying intricacies of the training process, such as the non-linear nature of the spikes [19] or the large computational cost of unfolding the network in time to perform backpropagation through time (BPTT) [20]. Several approaches have been proposed to solve this issue, including spike timing dependent plasticity [21], neoHebbian three factor learning [22], and fast learning without synaptic plasticity [23]. One strong candidate attracting attention is a biologically plausible learning algorithm called eligibility

propagation (e-prop) [24] that is based on the tag-and-capture hypothesis for synaptic adaptation [22]. The algorithm postulates that a chemical trace (e.g. Ca^{2+} ions) accumulates in the synapse in proportion to neural activity and that the network is then flooded by a neurotransmitter (e.g. dopamine or norepinephrine) that acts as a top-down learning signal [24]. Hence e-prop is naturally local (traces accumulate inside a neuron through activation alone) and online (information is propagated only forward in time), eliminating the need to unfold the network in time as in BPTT.

The combination of spiking nature of RSNNs and the local/online nature of e-prop gathered attention not only as a new biologically plausible machine learning but also for applications in neuromorphic hardware. For instance, the feasibility of implementing e-prop into neuromorphic hardware and the consequent advantages in terms of power consumption have been demonstrated through its implementation in SpiNNaker2 [25, 26] and in ASICs such as ReckOn [27]. However, in order to facilitate optimal hardware implementation, it is still imperative to analyze how network structures and input conditions affect learning performance and network dynamics. For instance, the neuronal connection probability in the brain is typically sparse, estimated at approximately 8%–20% in the mammalian cortex [28, 29], and mimicking such characteristics may further reduce wiring costs and power consumption in neuromorphic hardware. Additionally, biological networks are inherently robust to noise, a desirable feature in real-world applications of SNNs.

In this paper, we investigate the optimal configuration for e-prop by systematically analyzing the effects of network sparsity in the recurrent layer and evaluate the tolerance of the model to an injected random noise signal. As a benchmark, an RSNN with spiking input and continuous output is employed and trained to solve a complex temporal credit assignment task. The e-prop algorithm is used to train the input and recurrent layers while a standard gradient descent training is used for the output layer. We find that sparser networks create more diverse activation patterns in response to different inputs, leading to a significant improvement in task performance. Furthermore, the network also shows strong resilience to injected noise, which is found to be achieved by adapting the input layer to enhance the relevant signal component in the input channels while suppressing irrelevant information. This noise separation capability was found to be a characteristic feature of e-prop, distinguishing it from BPTT. These findings bolster the biological plausibility claims of the e-prop algorithm and provide a framework for optimizing network connectivity for classification tasks.

2. Methods

Neuron and network models

Replicating the experimental setup of the original e-prop paper [24], our RSNN comprises N_{rec} spiking neurons with a 50/50 split of two different types, leaky integrate-and-fire neurons (LIF) and adaptive LIF neurons (ALIF) [30, 31] (see supplementary materials figure S1, for the effect of LIF-ALIF ratio on task performance). In both neuron types, the behavior is governed by a real-valued quantity $v_i \in \mathbb{R}$ that we refer to as a membrane potential, borrowing the term from the biological analogue, which evolves according to the following equation:

$$v_i^{t+1} = \alpha v_i^t + \sum_j W_{ij}^{\text{in}} x_j^t + \sum_j W_{ij}^{\text{rec}} z_j^t - v_{\text{th}} z_i^t, \quad (1)$$

where $\alpha = e^{-\Delta t / \tau_{\text{mem}}}$ with τ_{mem} as the membrane time constant and Δt as the time-step, v_{th} is the firing threshold of the neuron, $x_j^t = \{0, 1\}$ and $z_j^t = \{0, 1\}$ are respectively the input signal and the observable state (spike) of neuron j at time t . $\mathbf{W}^{\text{in}} = [W_{ij}^{\text{in}}]$ and $\mathbf{W}^{\text{rec}} = [W_{ij}^{\text{rec}}]$ are synaptic weight matrices in the input and recurrent layers, respectively, and were initialized with a normalized Gaussian distribution $\mathcal{N}$ (see table 1).

Once v_i reaches a threshold value v_{th} , a spike is generated according to:

$$z_i^t = \begin{cases} \theta(v_i^t - v_{\text{th}}) & (\text{LIF}), \\ \theta(v_i^t - b_i^t) & (\text{ALIF}), \end{cases} \quad (2)$$

where θ is a Heaviside step function. For LIF neurons, v_{th} is a constant, while for ALIF neurons, the firing threshold $b_i \in \mathbb{R}$ is defined as:

$$b_i^t = v_{\text{th}} + \beta a_i^t \quad \text{with} \quad a_i^t = \rho a_i^{t-1} + z_i^{t-1}, \quad (3)$$

where $\rho = e^{-\Delta t / \tau_a}$, τ_a is the time constant for the adaptable threshold, and β is a heuristically defined constant [24].

After a spike, v_i is reset in the subsequent time step using the last term in equation (1).

Table 1. List of parameters used in the experiments.

Parameter	Value	Parameter	Value
f_0	4 Hz	Δt	1 ms
v_{th}	0.6	v_{rest}	0
τ_{mem}	20 ms	τ_a	2000 ms
β	$1.7(1 - \rho)/(1 - \alpha)$	N_{in}	40
N_{rec}	100 (50 LIF + 50 ALIF)	N_{out}	2
W_{ij}^{in}	$\mathcal{N}(0, 1)/\sqrt{N_{in}}$	W_{ij}^{rec}	$\mathcal{N}(0, 1)/\sqrt{N_{rec}}$
W_{ij}^{out}	$\mathcal{N}(0, 1)/\sqrt{N_{out}}$	B_{ij}^{out}	$\mathcal{N}(0, 1)/\sqrt{N_{out}}$
γ_d	0.3	η	0.005

Finally, the output of the network $\mathbf{y}^t = [y_k^t]$ is given by:

$$y_k^t = \sum_j W_{kj}^{out} z_j^t, \quad (4)$$

where $\mathbf{W}^{out} = [W_{kj}^{out}]$ is the readout weight matrix, and $k = (1, \dots, K)$ is the index for the classification category.

The three layers are initialized as a fully connected matrix, some synapses of the recurrent layer are then pruned to implement sparsity. These synapses are chosen differently in each iteration of the experiment but are kept fixed throughout each training process.

The values of each parameter have been set to reproduce those in the original e-prop study by Bellec *et al* [24] and have been kept fixed unless otherwise noted. They are summarized in table 1.

Task

In the complex temporal credit assignment task, an agent (a rodent in the original studies) is exposed to cues from left and right and needs to decide which side to take after a delay period [32, 33]. The complexity of the task arises from the long delay between stimuli and reward, as well as the requirement for the agent to autonomously infer the logic linking the cues to the correct answer given by the reward signal. The input is given across N_{in} neurons as Poisson spikes [34] with frequency f_0 divided among four channels as: left and right stimuli, the reward signal, and background noise.

Each episode comprises seven randomized input packets from left and right, followed by a delay period, and ends with the reward packet giving the correct answer to the agent. The probability of an input being in the left/right channel is set to 30%/70% to reproduce the experimental setup of the original e-prop study [24]. Each packet lasts for 150 ms, the delay period is 1050 ms thus the total length of an episode amounts to 2250 ms (see supplementary materials figure S2, for the effect of cue distribution on task performance). This is a classification task with $K = 2$ categories, so we define the loss function E as the cross-entropy error in the following equation:

$$E = - \sum_{t,k} \pi_k^{*,t} \log \pi_k^t, \quad (5)$$

where $\pi_k^{*,t}$ is a one-hot encoded K -dimensional target vector given as the reward packet, and $\pi_k^t = \frac{\exp(y_k)}{\sum_{k'} \exp(y_{k'})}$.

During each epoch of training, the network is exposed to 64 randomly generated episodes. Testing is performed on 16 batches each containing 64 generated episodes. Unless otherwise noted, the noise-to-signal ratio was set to 50%, such that the probability of a spike being generated in the noise channel is half of the probability of generating a spike in a signal channel.

Network evolution

We trained both $\mathbf{W}^{in}$ and $\mathbf{W}^{rec}$ using e-prop since all information the input and recurrent layers handle is in the form of spikes. The output signal, on the other hand, is continuous thus requiring a different learning algorithm for training $\mathbf{W}^{out}$; in this work, we employed a simple gradient descent method:

$$\Delta W_{ij}^{out} = \frac{dE}{dW_{ij}^{out}} = \sum_t (\pi_i^t - \pi_i^{t,*}) z_j^t. \quad (6)$$

The e-prop algorithm is based on gradient descent:

$$\Delta W_{ij} = \frac{dE}{dW_{ij}} = \eta \sum_t \frac{dE}{dz_i^t} \left[\frac{dz_i^t}{dW_{ij}} \right]_{local}, \quad (7)$$

where η is the learning rate (table 1). The term $\frac{dE}{dz_i^t}$ represents the learning signal L , which depends on the definition of the loss function. For the case at hand, L for neuron i at time t takes the form:

$$L_i^t := \frac{dE}{dz_i^t} \approx \frac{\partial E}{\partial z_i^t} := \sum_k B_{ik}^{\text{out}} (\pi_k^t - \pi_k^{t,*}). \quad (8)$$

Here, we used an approximation from total to partial derivatives to preserve the online nature of the algorithm. In particular, using the total derivative would entail having information on the impact of z_i^t on E generated by future spikes of different neurons, which is not compatible with online information processing. This approximation involves substituting the output matrix $\mathbf{W}^{\text{out}}$ with a random fixed matrix $\mathbf{B}^{\text{out}} = [B_{ik}^{\text{out}}]$, drawing inspiration from the broadcast alignment method [35].

The local term on the RHS of equation (7) is the eligibility trace (e_{ij}), defined as:

$$e_{ij}^t := \left[\frac{dz_i^t}{dW_{ij}} \right]_{\text{local}} := \frac{\partial z_i^t}{\partial \mathbf{h}_i^t} \sum_{t' \leq t} \frac{\partial \mathbf{h}_i^t}{\partial \mathbf{h}_i^{t-1}} \cdots \frac{\partial \mathbf{h}_i^{t'+1}}{\partial \mathbf{h}_i^{t'}} \frac{\partial \mathbf{h}_i^{t'}}{\partial W_{ij}} := \frac{\partial z_i^t}{\partial \mathbf{h}_i^t} \epsilon_{ij}^t, \quad (9)$$

where $\mathbf{h}_i^t = [v_i^t, b_i^t]$ is the internal state of the network that evolves according to equations (1) and (3) to generate spiking activity in the visible state according to equation (2). The summed term on the RHS is labeled as the eligibility vector ϵ_{ij} and can be computed recursively according to:

$$\epsilon_{ij}^t = \frac{\partial \mathbf{h}_i^t}{\partial \mathbf{h}_i^{t-1}} \epsilon_{ij}^{t-1} + \frac{\partial \mathbf{h}_i^t}{\partial W_{ij}}. \quad (10)$$

The spike function z_i^t is non-differentiable, so its derivatives have been approximated by a tent function:

$$\psi_i^t = \frac{1}{v_{\text{th}}} \gamma_d \max \left(0, 1 - \left| \frac{v_i^t - b_i^t}{v_{\text{th}}} \right| \right), \quad (11)$$

where γ_d is a constant. However, for the sake of clarity, we have written the derivatives of z_i^t using the standard derivative symbol in a slight abuse of notation.

A more in-depth derivation of the traces for both the LIF and ALIF neurons can be found in the supplementary materials and in [24].

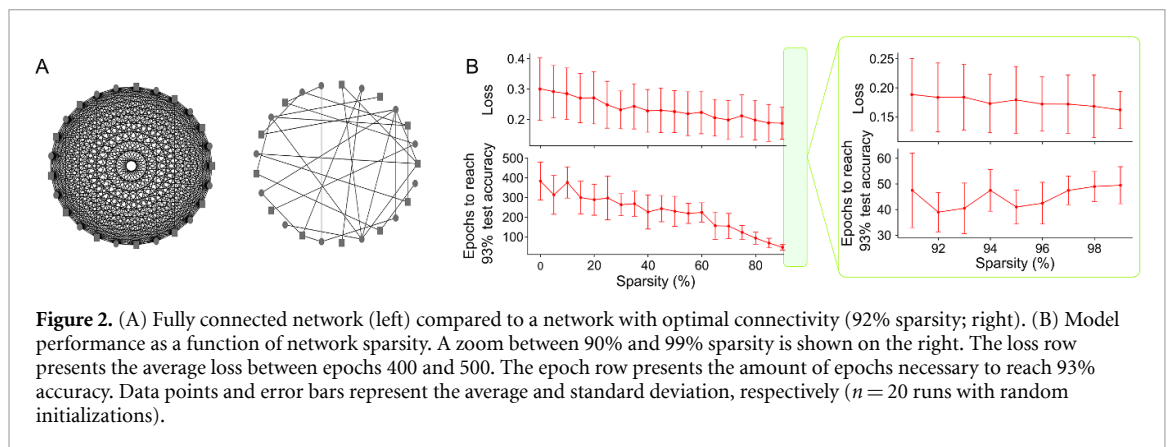
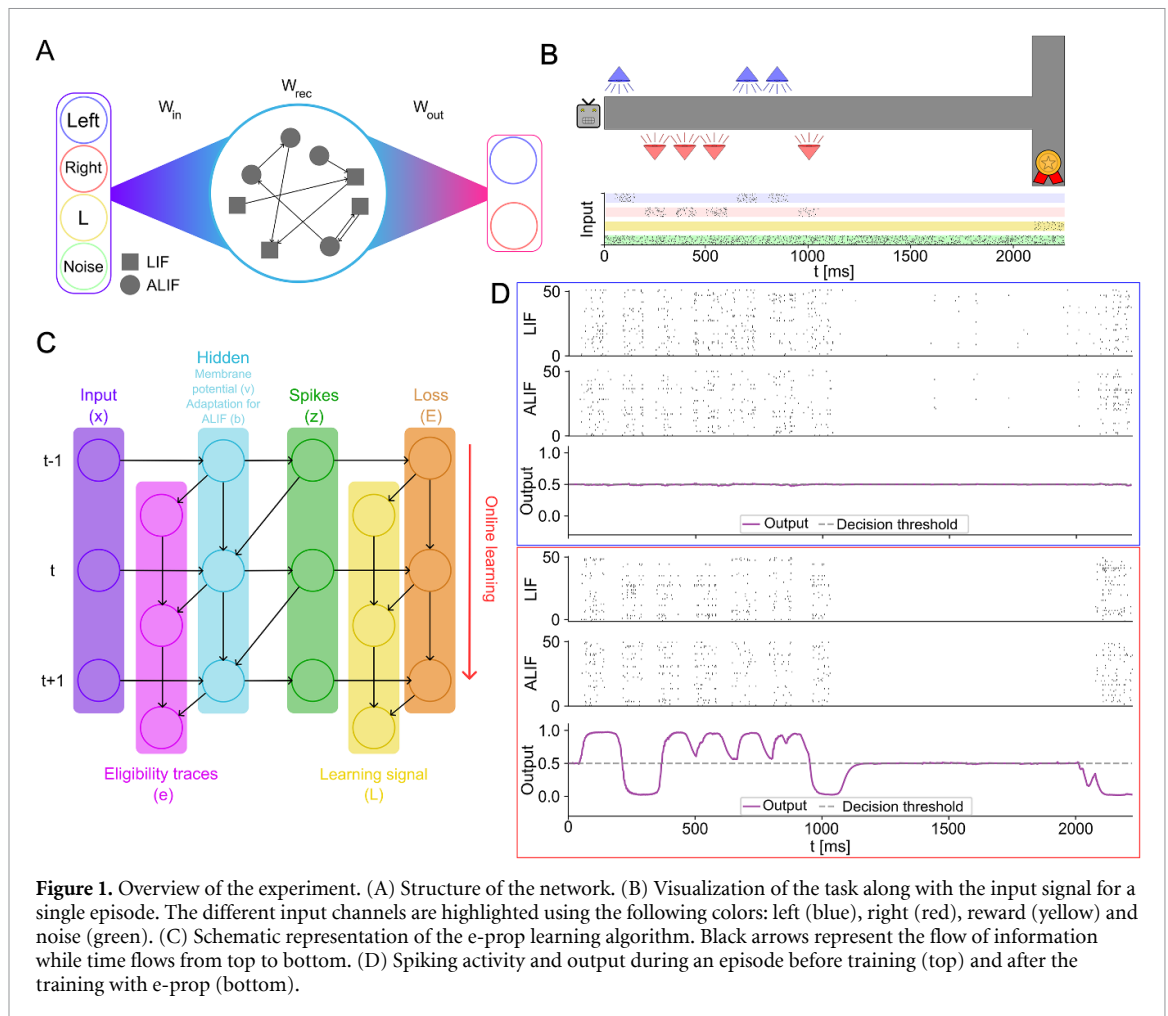
3. Results

Our RSNN model is represented in figure 1(A). In this section, we analyze the behavior of the model in solving the complex temporal credit assignment task (figure 1(B)). In particular, we studied the impact of connectivity on the e-prop learning algorithm as well as the network's response to various amounts of separable input noise. We also analyzed the impact of learning on the landscape of synaptic connections. Finally, we compared our results with a BPTT-trained network. A schematic representation of the e-prop algorithm, along with an overview of the network's dynamics and output before and after training, is presented in figures 1(C) and (D).

A biologically realistic connectivity improves e-prop training performance

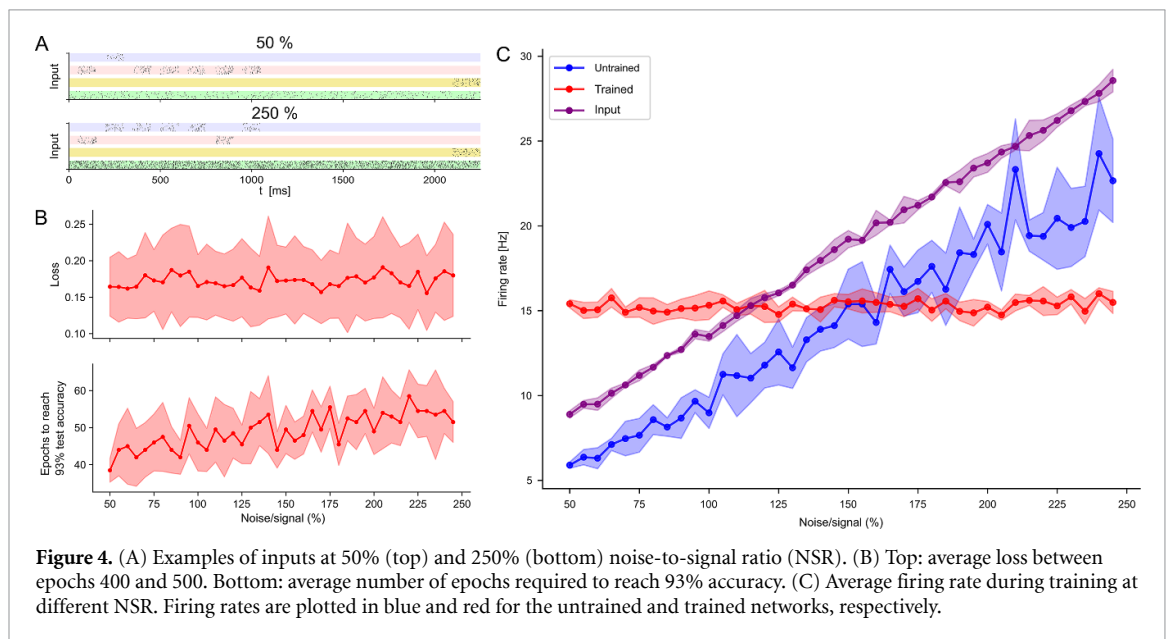
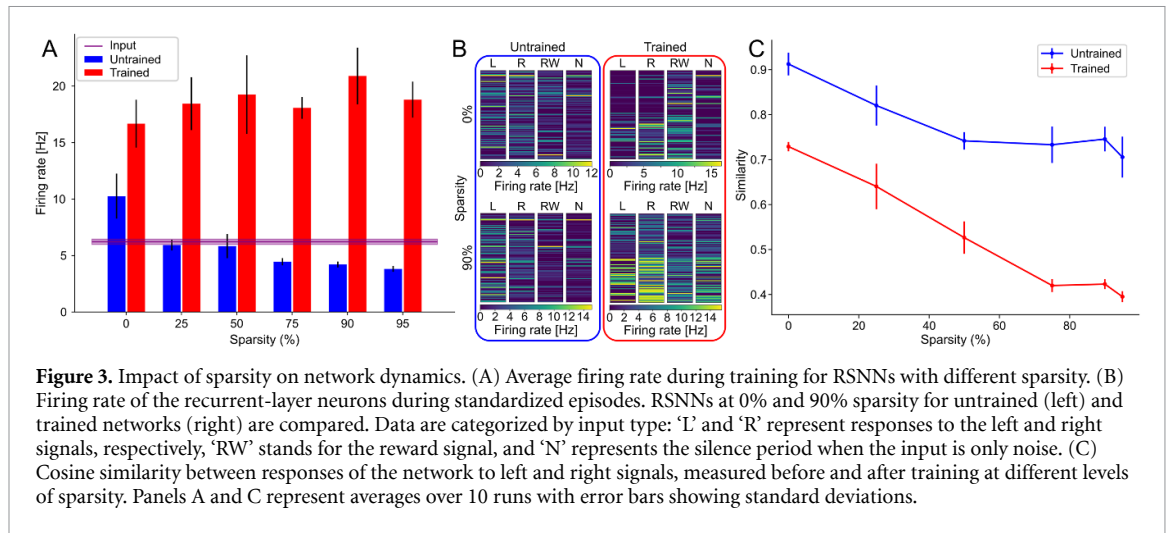
We first investigated how network connectivity impacts the behavior and performance of e-prop trained RSNNs. We focused on connection density, since neurons in the nervous system of animals are generally sparsely connected [28, 29, 36]. We started with a fully-connected network and pruned different randomly selected synapses (figure 2(A)). As summarized in figure 2(B), networks with a lower connection density exhibited superior performance both in terms of loss after learning (loss row) and in terms of training efficiency (epoch row). Notably, we found that the optimal sparsity was at 92% (i.e. connection probability of 8%), showing the most efficient training in terms of epochs to achieve the accuracy threshold (set to 93% [24]). The average loss observed at that sparsity (0.18 ± 0.06 , $n = 20$ samples) was not statistically different from the minimum average loss obtained at a sparsity of 99% (0.16 ± 0.03 , $n = 20$ samples; two-sided t -test, $p > 0.05$). This connection density closely resembles that reported in murine studies [28], showing that a more biologically realistic network improves the performance of e-prop. The advantage of sparser networks over fully connected ones at solving the complex credit assignment task was robust to changes in recurrent layer size (supplementary materials figure S3).

To further substantiate these findings, we investigated the networks' responses to a standardized input at different sparsity levels. Analyses revealed that while the average firing rate of the untrained network decreased with increasing sparsity, the training process regularized the network's overall activity, such that



the average firing rate converged to a constant value independent of the connection density (figure 3(A)). This value was found to be approximately 16 Hz, which was determined by the balance between the classification loss minimization and a regularization term for the firing rate (see supplementary materials, section ‘Firing rate regularization’).

The effect of training was also apparent on network responses to different input signals (figure 3(B)). Comparing the cosine similarity between the network’s responses to the left and right input channels, we observed a consistent reduction in similarity post-training (blue vs. red curves), indicating an enhanced capacity for input discrimination (figure 3(C)). Furthermore, the similarity decreased with increasing sparsity, suggesting that more sparsely connected networks are intrinsically more effective at differentiating between input signals. This ability to generate distinct activation patterns may underlie the improved performance observed at higher levels of sparsity.

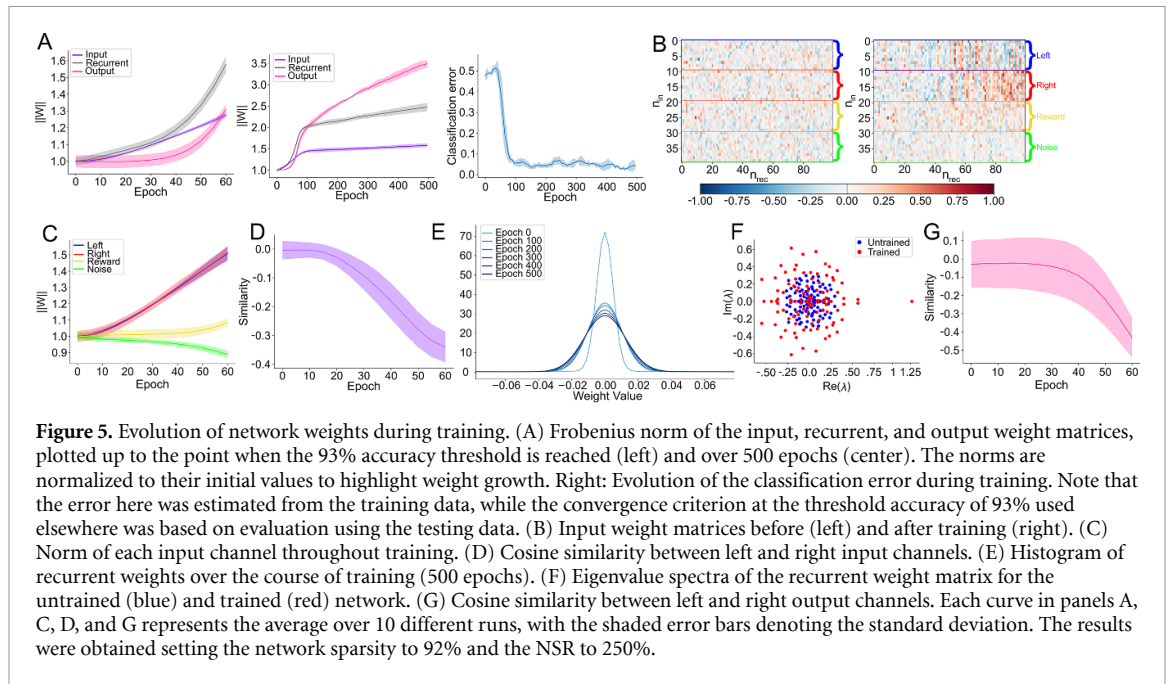


To assess the generality of our findings, we tested how network sparsity affects the performance of the e-prop training algorithm on another classification task using the TIMIT dataset [24, 37]. This task was chosen since, while still being a classification task, it involves (1) a larger number of output classes, (2) no explicit pattern delay or silence periods, and (3) variable input patterns based on natural speech cadence and content, making the task a complementary test case. We confirmed that sparsity improves the performance, with the optimal value (94%) compatible with the one found for the complex temporal credit assignment task (92%). Full details about the experiment and results are provided in the supplementary materials, section 'Phoneme recognition task' and figure S4.

E-prop-trained input layer separates noise

Next, we studied how different levels of input noise affected the performance of the e-prop algorithm. To control the level of background activity in RSNNs, we modulated the frequency of the Poisson process that generates input spikes in the noise channel. Some examples of the inputs employed during the experiment are reported in figure 4(A). The input noise we employed in this experiment is asynchronous and uncorrelated, which resembles thalamic spontaneous activity [38]. Network sparsity was kept fixed at the optimal value obtained in the previous section, namely 92%.

Figure 4(B) shows the network's change in behavior with the increase of noise. We see that the epochs required to reach the desired accuracy of 93% increased from an average of 38.5 ± 3.2 at 50% noise-to-signal ratio (NSR) to 51.5 ± 5.5 at 250% NSR ($n = 20$ samples; two-sided t -test, $p < 0.01$); however, the change stayed within 2σ and the network was still capable of solving the task at very high noise levels. While the loss



showed a modest change from 0.16 ± 0.04 at 50% NSR to 0.18 ± 0.06 at 250% NSR, this change was not statistically significant (two-sided t -test, $p > 0.05$).

To further characterize the network activity under various noise intensity, we evaluated the mean firing rate of the recurrent layer at different stages of training and analyzed its behavior in response to a standardized input with different amounts of noise. As summarized in figure 4(C), the untrained network's activity grows proportionally to the density of input spikes, while the trained network consistently maintained a firing rate of approximately 16 Hz, independent of the noise injected. This result extends the data from the previous section, showing that the network also converges towards a stable firing rate independently of the amount of noise injected. The ability of the network to solve the complex temporal credit assignment task, even under high NSR conditions, suggests an intrinsic mechanism for noise separation in the input layer that stabilizes network activity in the recurrent layer. Notably, when the random noise was superimposed on the input channel, task performance significantly decreased as noise levels were increased; however, e-prop remained capable of solving the task even under substantial noise, up to 15% NSR (see supplementary materials, section 'Noise filtering' and figure S5 for details).

Network mechanisms of e-prop training

We then examined the network's evolution during training to better understand how the synaptic landscape is shaped by training. Analysis revealed that all three layers evolved through time and thus contributed to learning (figure 5(A)).

We also verified that changes in the input and recurrent weight matrices, both trained under e-prop, plateaued at approximately 60 epochs around which the accuracy threshold of 93% was met (figure 5(A), center). These results confirm that e-prop converged to an optimal state for solving the task at hand. While the norm of the readout weight matrix did not saturate within 500 epochs, perhaps reflecting the typical convergence characteristics of the standard gradient descent, it had only a mild impact on the accuracy (figure 5(A), right).

In the input layer, each input channel exhibited distinct patterns of evolution (figure 5(B)). In particular, as suggested in the previous section, the layer autonomously learned to separate noise and highlight input signals. As can be seen in figure 5(C), the norm of the signal channels increased, while that of the noise channel was suppressed.

To gain further insight into how the left and right channels evolved during the training process, we computed the cosine similarity between their respective weights over time (figure 5(D)). Analysis revealed a pronounced decrease in similarity, indicating that the input layer evolved to route the two input channels to distinct neuron populations of the recurrent layer. This divergence is consistent with the distinct activation patterns observed in previous sections. Similar mechanisms have been observed in the sensory complex in the brain. In particular, the thalamus has been shown to adaptively filter inputs in response to feedback from

the cortex [39] and to route information to bootstrap learning [40, 41]. These findings here explain the network's resilience to injected noise and its enhanced performance at higher sparsity.

Analysis of the recurrent layer revealed a pronounced widening the weight distribution throughout training, while roughly maintaining the initial balance of excitation and inhibition (figure 5(E)). Furthermore, the eigenvalue spectrum of the recurrent weight matrix broadened substantially, with the spectral radius increasing up to a value of 1.25 (figure 5(F)). This behavior of the spectral radius is usually associated with a system approaching the 'edge of chaos', a regime of operation characterized by enhanced temporal stability and memory capacity [42, 43].

Lastly, we evaluated the evolution of the output layer. Consistent with the input layer, the two output channels evolved their synaptic weights to readout from different neurons in the recurrent layer. This differentiation is reflected in a decrease in cosine similarity between the left and right output channels (figure 5(G)), supporting our hypothesis that the network learns to route distinct neuron populations in the recurrent layer to each output channel.

Comparison with BPTT

In order to better contextualize our results in the landscape of RSNN training, we performed a comparative analysis between e-prop and BPTT. BPTT entails unraveling the network through time, considering each time step as the layer of a feed-forward deep neural network. After the forward pass, the loss function is calculated for each layer and backpropagation is used to compute the gradients, which are then used to update the weights [20]. For all simulations presented in this section, the learning rates of the two algorithms were optimized to maximize the training efficiency ($\eta = 0.02$, for both e-prop and BPTT), to ensure a fair comparison.

We first analyzed the performance of a fully connected network using the standard configuration employed throughout this paper and found that BPTT was faster and more efficient at solving the complex temporal credit assignment task. While the performance gap between the two learning algorithms remained statistically significant throughout all noise levels (two-sided t -test, $p < 0.05$, $N = 10$), the gap decreased when they were evaluated at their respective optimal network sparsity (90% for BPTT and 92% for e-prop), (figure 6(A)).

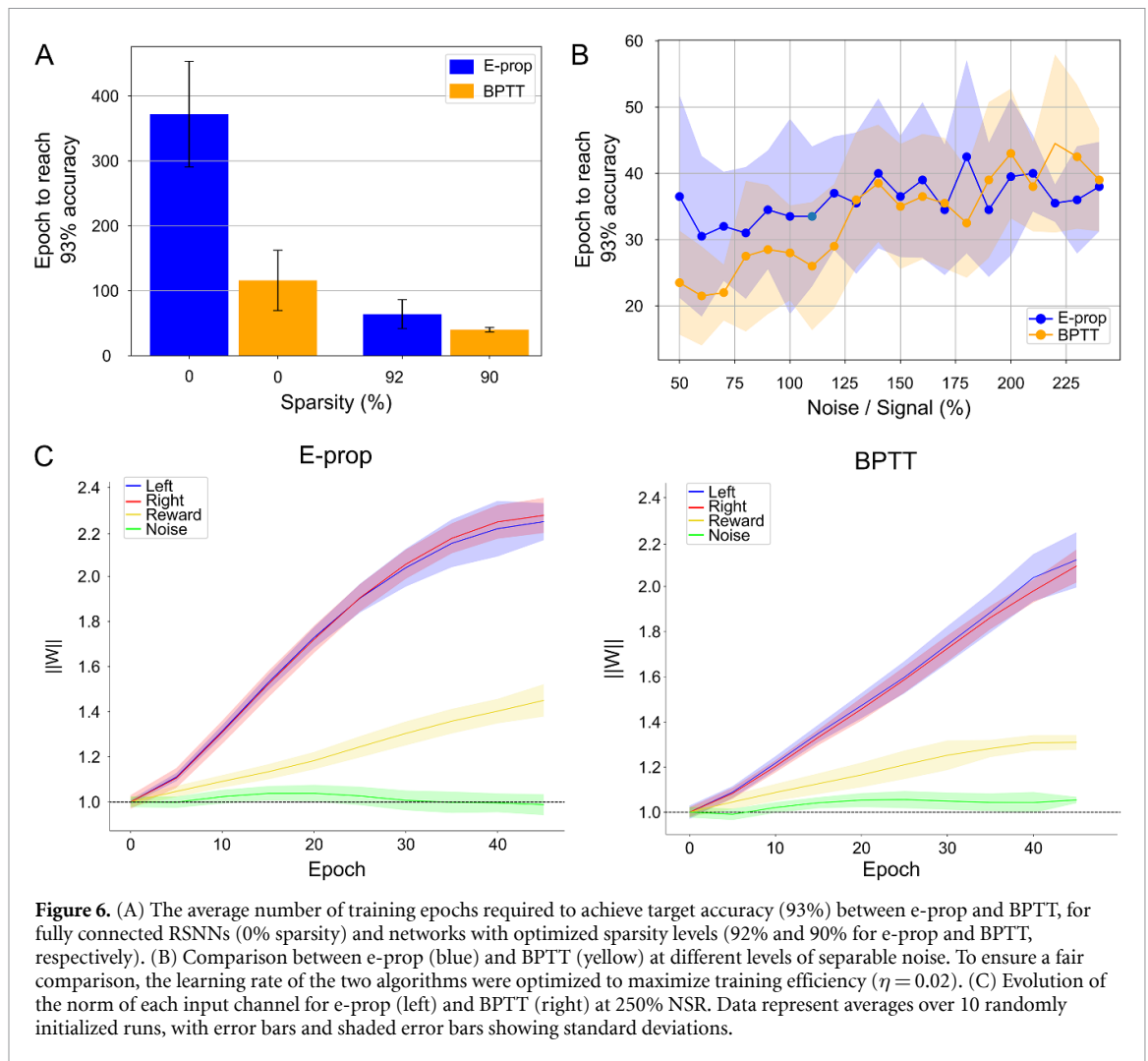
Interestingly, the performance of BPTT significantly decreased with increasing noise level, reaching a level statistically indistinguishable from e-prop above 70% NSR (two-sided t -test, $p > 0.05$, $N = 10$) (figure 6(B)). In contrast, e-prop maintained stable performance across various noise levels. Further investigation on the underlying mechanisms revealed that e-prop suppressed the contribution from the noise channel in the input weights (green line), while BPTT amplified the noise input over training epochs (figure 6(C)). This suggests that the noise separation capability was a property of the e-prop algorithm itself, rather than being a mere consequence of supervised learning (see supplementary materials, section 'Detailed comparison of e-prop and BPTT' and figure S6 for further details).

4. Discussion

In this study, we characterized the behavior of a RSNN trained using e-prop at different connection density and strengths of input noise. Simulations revealed that sparser networks are more effective at translating different inputs into distinct activation patterns in the recurrent layer which makes them better at solving the complex temporal credit assignment task. It was also found that the e-prop trained input layer adapts to act as a filter and increases the networks' resilience to injected noise, while such a property was absent in the BPTT-trained one.

Building on the biological plausibility claim of e-prop [24], we delved deeper into the learning mechanisms by demonstrating a thalamus-like functionality in the input layer and showing improved performance in networks with more biologically-plausible connectivity. The results of this paper reinforce the evidence found in [44] about the resemblance between the input layer and the thalamus in suppressing irrelevant input information such as noise. Our findings also extend previous work on the role of sparsity in ANNs [45, 46] by demonstrating similar benefits in RSNNs trained with a biologically plausible algorithm. Additionally, the results confirm the earlier findings on the increased efficacy of sparser RSNNs in solving classification tasks [47], generalizing the conclusion to tasks with long temporal delays between inputs and rewards.

Results herein lay down a basis for optimization of RSNN morphology to solve classification tasks. Future work may explore more complicated network structures, such as the modular structure we explored in the past using biological neurons [48, 49]. It will also be interesting to explore other means to simulate spontaneous activity, background activity that occurs even in the absence of an input signal, and originates from various sources [50]. Although this property is typically disregarded in conventional artificial neural



networks for machine learning, it constitutes a fundamental aspect of biological neural networks. There are multiple theories about the role this spontaneous activity plays in a plethora of neural processes, such as contributing to consolidation and retrieval in memory [51], modulating the integration capabilities of neurons in sensory tasks [52] or introducing variability to computation in cognitive and sensory applications, similarly to noise in machine learning [53, 54]. Further studies could test the impact of connectivity and noise on unsupervised training and reinforcement learning, as well as apply the e-prop learning algorithm to a biologically inspired computational paradigm as in [55].

The features of the e-prop algorithm we highlighted are promising for its implementation in neuromorphic hardware: in particular, the intrinsic mechanism that minimizes the impact of irrelevant input, as well as the sparse connectivity found to be optimal, could significantly alleviate the hardware requirements of the implementation and simplify the design process for an ASIC such as ReckOn [27]. While further studies employing more diverse and complex datasets are needed to fully validate the generalizability of our conclusions to other tasks, prior work in RSNN suggests that the impact of network topology on performance is mostly task-independent [56, 57]. These studies would advance the application of biologically inspired approaches in neuromorphic systems, while simultaneously providing insights into the functional significance of biological architectures shaped by evolution.

Data availability statement

This study forms part of a broader ongoing research project; consequently, the data cannot be disclosed publicly at this time. The data that support the findings of this study are available upon reasonable request to the authors.

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Conflict of interest

The authors declare that there are no conflicts of interest.

Author contributions

Davide Noè

Conceptualization (equal), Data curation (lead), Formal analysis (lead), Investigation (lead), Methodology (lead), Resources (lead), Software (lead), Validation (lead), Visualization (lead), Writing – original draft (equal)

Hideaki Yamamoto  0000-0003-3362-5376

Conceptualization (equal), Funding acquisition (equal), Project administration (lead), Supervision (equal), Writing – original draft (equal)

Yuichi Katori  0000-0003-2773-0786

Funding acquisition (supporting), Project administration (supporting), Supervision (supporting), Writing – review & editing (equal)

Shigeo Sato  0000-0003-3912-357X

Funding acquisition (supporting), Project administration (supporting), Supervision (equal), Writing – review & editing (equal)

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